MTH318

Timed Online Assignment – January Semester 2022

Advanced Calculus

Friday, 27 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Define a sequence of functions $\{f_n(x)\}$ on the interval $[0,1]$, where

$$f_n(x) = \frac{x^2}{x^2(1+n^2) - 2nx + 1}.$$

(a) Give a brief explanation why $f_n(x)$ is continuous on $[0,1]$ for each integer n.
(3 marks)

(b) Show that $\{f_n(x)\}$ is uniformly bounded on $[0,1]$, that is, there exists some positive real number M such that $|f_n(x)| \leq M$ for all $n \geq 1$ and for all $x \in [0,1]$.
(4 marks)

(c) Show that $\{f_n(x)\}$ converges pointwise to a continuous function $f(x)$ on $[0,1]$, but that the convergence is not uniform.
(8 marks)

(d) Prove that there exists some positive real number ε satisfying the following condition: For any $\delta > 0$ and for any positive integer N, there exists some integer $n \geq N$ and some $x, y \in [0,1]$ such that $|x - y| < \delta$ and $|f_n(x) - f_n(y)| \geq \varepsilon$.
(10 marks)

Question 2

(a) Determine whether each of the infinite series is absolutely convergent, conditionally convergent, or divergent. Justify your answers fully.

(i)

$$\sum_{n=1}^{\infty} \frac{\sqrt{3}\cos(n) + \sin(n) + 3}{3n + 2}$$

(7 marks)

(ii)

$$\sum_{n=1}^{\infty} \left(\frac{1 - 2n}{3n + 5}\right)^n$$

(5 marks)

(b) Suppose that $\{a_n\}$ is a sequence of positive real numbers such that the series

$$\sum_{n=1}^{\infty} a_n$$

is convergent. Prove that the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$$

is also convergent.

(8 marks)

Question 3

(a) Show that the series of functions

$$\sum_{n=1}^{\infty} \frac{x\ln(x)}{1+x^2n^2}$$

converges uniformly on the interval $[1, \infty)$.

(8 marks)

(b) Show that for any given positive integer N,

$$\left|\sum_{n=1}^{N} (-1)^n \sin(x)\right| \leq |\sin(x)|$$

for all $x \in [0, 2\pi]$.

(3 marks)

(c) Show that the series of functions

$$\sum_{n=1}^{\infty} \frac{(-1)^n x^2 \sin(x)}{\sqrt{n}}$$

converges uniformly on the interval $[0, 2\pi]$.

(5 marks)

(d) Discuss whether there is uniform convergence of the series of functions

$$\sum_{n=1}^{\infty} \frac{(-1)^n x^2 \sin(x)}{\sqrt{n}}$$

on the unbounded interval $[0, \infty)$.

(4 marks)

Question 4

(a) Compute the radius of convergence and the interval of convergence of the power series

$$\sum_{n=0}^{\infty}(-1)^n \frac{2\cdot 4\cdot 6\cdots(2n)\cdot(2n+2)}{1\cdot 3\cdot 5\cdots(2n-1)\cdot(2n+1)} x^{n+1}.$$

(7 marks)

(b) Use the fact that

$$\tan^{-1}(x^2) = \int_0^x \frac{2t}{1+t^4}\, dt \quad (x \in \mathbb{R})$$

to derive a power series expansion for $\tan^{-1}(x^2)$ about the point $x = 0$, and find the interval of convergence.

(7 marks)

(c) Find a function that is infinitely differentiable at $x = 1$ and has a Taylor series expansion about $x = 1$ given by

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{n\cdot 3^n}.$$

(6 marks)

Question 5

(a) Suppose that $\{f_n(x)\}$ is a sequence of continuous functions on an interval D such that $\{f_n(x)\}$ converges uniformly to a continuous function $f(x)$ on D. Show that whenever $\{x_n\}$ is a sequence in D that converges to $x \in D$, then

$$\lim_{n\to\infty} f_n(x_n) = f(x).$$

(9 marks)

(b) Let

$$g_n(x) = \frac{1}{1+nx}$$

for all $x \in (0,1)$. Show that $\{g_n(x)\}$ is a sequence of continuous functions on $(0,1)$ that converges pointwise to the zero function on $(0,1)$, and

$$\lim_{n\to\infty} g_n(x_n) = 0$$

whenever $\{x_n\}$ is a sequence in $(0,1)$ that converges to $x \in (0,1)$, but the convergence of $\{g_n(x)\}$ to the zero function is not uniform on $(0,1)$.

(6 marks)

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